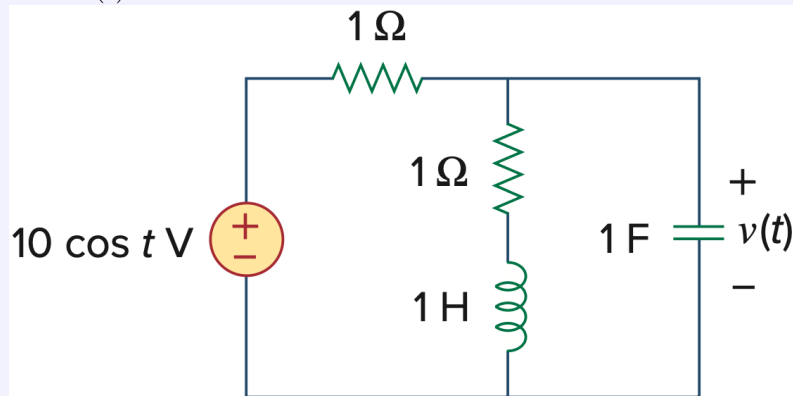


Exercise Sheet #8: AC Circuits

Discussion: Dec 8, 2025

Exercise 1: *RLC* CircuitFind $v(t)$ in the circuit below.

Solution

We are given the time-domain source

$$v_s(t) = 10 \cos t \text{ V}.$$

In complex (exponential) representation we write this as the real part of a complex exponential:

$$v_s(t) = \operatorname{Re}(\hat{v}_s e^{j\omega t}), \quad \hat{v}_s = 10, \quad \omega = 1 \text{ rad/s},$$

$$\text{so } v_s(t) = \operatorname{Re}(10e^{jt}).$$

At $\omega = 1$,

$$\underline{Z}_{R_1} = 1, \quad \underline{Z}_{R_2} = 1, \quad \underline{Z}_L = j\omega L = j, \quad \underline{Z}_C = \frac{1}{j\omega C} = -j.$$

The series $R_2 - L$ branch impedance is

$$\underline{Z}_{RL} = 1 + j.$$

The parallel combination of $\underline{Z}_{RL}$ and $\underline{Z}_C$ has admittance

$$\underline{Y}_p = \frac{1}{\underline{Z}_{RL}} + \frac{1}{\underline{Z}_C} = \frac{1}{1+j} + \frac{1}{-j}.$$

Compute each term explicitly:

$$\frac{1}{1+j} = \frac{1-j}{(1+j)(1-j)} = \frac{1-j}{2} = \frac{1}{2} - \frac{1}{2}j, \quad \frac{1}{-j} = j.$$

Thus

$$\underline{Y}_p = \left(\frac{1}{2} - \frac{1}{2}j\right) + j = \frac{1}{2} + \frac{1}{2}j.$$

Therefore the equivalent impedance of the parallel network is

$$\underline{Z}_p = \frac{1}{\underline{Y}_p} = \frac{1}{\frac{1}{2} + \frac{1}{2}j} = \frac{\frac{1}{2} - \frac{1}{2}j}{(\frac{1}{2})^2 + (\frac{1}{2})^2} = 1 - j.$$

The total series impedance seen by the source is

$$\underline{Z}_{\text{tot}} = \underline{Z}_{R_1} + \underline{Z}_p = 1 + (1 - j) = 2 - j.$$

Now form the complex current $\hat{i}$ (the complex amplitude multiplying e^{jt}) by dividing the complex source amplitude by the complex impedance:

$$\hat{i} = \frac{\hat{v}_s}{\underline{Z}_{\text{tot}}} = \frac{10}{2 - j}.$$

Perform the algebraic division by multiplying numerator and denominator by the conjugate:

$$\hat{i} = 10 \frac{2 + j}{(2 - j)(2 + j)} = 10 \frac{2 + j}{5} = 4 + 2j.$$

Thus the complex current amplitude is $\hat{i} = 4 + 2j$. (Its magnitude is $|\hat{i}| = \sqrt{4^2 + 2^2} \approx 4.472$ and its angle $\arg(\hat{i}) = \arctan(2/4) = 26.57^\circ$, if one prefers polar form.)

The complex voltage across the parallel network (and thus across the capacitor) is

$$\hat{v}_p = \hat{i} \underline{Z}_p = (4 + 2j)(1 - j).$$

Multiply out:

$$\hat{v}_p = 4(1 - j) + 2j(1 - j) = 4 - 4j + 2j - 2j^2 = 4 - 2j + 2 = 6 - 2j.$$

So the complex amplitude of the capacitor voltage is $\hat{v}_p = 6 - 2j$.

We may express this two ways:

(a) Time-domain real part of the exponential:

$$v(t) = \text{Re}(\hat{v}_p e^{jt}) = \text{Re}((6 - 2j)e^{jt}).$$

Expanding $(6 - 2j)e^{jt} = (6 - 2j)(\cos t + j \sin t)$ and taking the real part gives

$$v(t) = 6 \cos t + 2 \sin t.$$

(b) Amplitude–phase form: compute magnitude and angle of $\hat{v}_p$:

$$|\hat{v}_p| = \sqrt{6^2 + (-2)^2} \approx 6.325,$$

$$\arg(\hat{v}_p) = \arctan\left(\frac{-2}{6}\right) \approx -18.43^\circ.$$

Thus

$$\hat{v}_p = 6.325 \angle (-18.43^\circ),$$

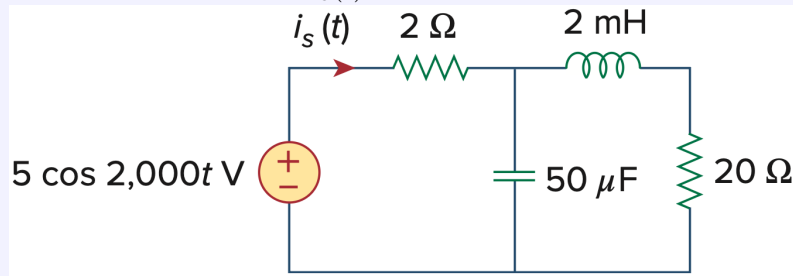
and therefore

$$v(t) = 6.325 \cos(t - 18.43^\circ) \text{ V},$$

which is algebraically equivalent to $v(t) = 6 \cos t + 2 \sin t$.

Exercise 2: Phase-shifted Source Current

Determine the value of $i_s(t)$ in the circuit below.

**Solution**

$$\omega = 2000 \text{ rad/s}$$

$$\underline{Z}_C = \frac{1}{j\omega C} = \frac{-j}{2000 \cdot 50 \times 10^{-6}} \Omega = -j10 \Omega$$

$$\underline{Z}_{RL} = 20 + j\omega L = (20 + j2000 \cdot 2 \times 10^{-3}) \Omega = (20 + j4) \Omega$$

$$\underline{Y}_p = \frac{1}{\underline{Z}_C} + \frac{1}{\underline{Z}_{RL}} = j0.1 + \frac{1}{20 + j4} = j0.1 + \frac{20 - j4}{416} = \left(\frac{20}{416} + j\frac{37.6}{416} \right) \text{ S}$$

$$\underline{Z}_p = \frac{1}{\underline{Y}_p} = \left(\frac{\frac{20}{416} - j\frac{37.6}{416}}{\left(\frac{20}{416}\right)^2 + \left(\frac{37.6}{416}\right)^2} \right) \Omega = (4.58716 - j8.62385) \Omega$$

$$\underline{Z}_{\text{tot}} = \underline{Z}_{R_2} + \underline{Z}_p = 2 \Omega + (4.58716 - j8.62385) \Omega = (6.58716 - j8.62385) \Omega$$

$$\hat{i}_s = \frac{\hat{v}_s}{\underline{Z}_{\text{tot}}} = \frac{5}{6.58716 - j8.62385} \text{ A} = (0.27714 + j0.36949) \text{ A}$$

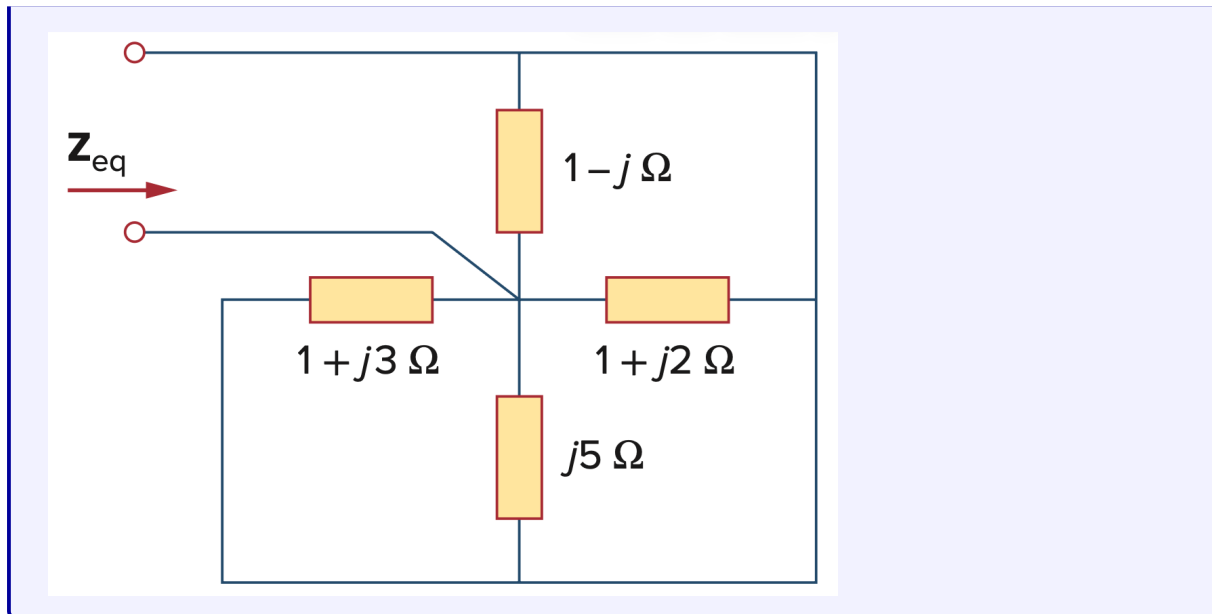
$$\hat{i}_s = \sqrt{(0.27714)^2 + (0.36949)^2} \text{ A} = 0.460753 \text{ A}$$

$$\varphi_{i_s} = \tan^{-1}\left(\frac{0.36949}{0.27714}\right) = 52.6263^\circ$$

$$i_s(t) = 460.7 \cos(2000t + 52.63^\circ) \text{ mA}$$

Exercise 3: Equivalent Impedance

Find $\underline{Z}_{eq}$ in the circuit below.



Solution

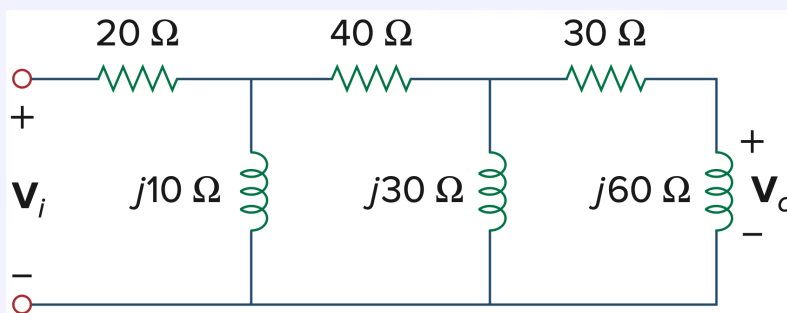
$$\underline{Y} = \frac{1}{\underline{Z}_1} + \frac{1}{\underline{Z}_2} + \frac{1}{\underline{Z}_3} + \frac{1}{\underline{Z}_4} = \frac{1}{1-j} + \frac{1}{1+j2} + \frac{1}{1+j3} + \frac{1}{j5}$$

$$= \frac{1+j}{2} + \frac{1-j2}{5} + \frac{1-j3}{10} + \frac{-j5}{25} = \frac{25+10+5+j(25-20-15-10)}{50} = (0.8 - j0.4) \text{ S}$$

$$\underline{Z}_{eq} = \frac{1}{\underline{Y}} = \frac{1}{0.8 - j0.4} = \frac{0.8 + j0.4}{0.8^2 + 0.4^2} = (1 + j0.5) \Omega$$

Exercise 4: Stepwise Voltage Division

- Calculate the phase shift of the circuit below.
- State whether the phase shift is leading or lagging (output with respect to input).
- Determine the magnitude of the output when the input is 120 V.



Solution

- We take the input $\underline{v}_i$ as the reference phasor (angle 0°) and denote the node voltages by $\underline{v}_1, \underline{v}_2, \underline{v}_3$ (with $\underline{v}_3 = \underline{v}_o$).

Step 1: Relation between $\underline{v}_3$ and $\underline{v}_2$

Node 3 is a simple divider between the series resistor R_3 and the inductor $\underline{Z}_3$:

$$\underline{v}_3 = \underline{v}_2 \cdot \frac{\underline{Z}_3}{R_3 + \underline{Z}_3}.$$

With $R_3 = 30$, $\underline{Z}_3 = j60$:

$$\begin{aligned} \frac{\underline{v}_3}{\underline{v}_2} &= \frac{j60}{30 + j60} = \frac{1}{1 - j0.5} = 0.8 + 0.4j \\ &= \sqrt{0.8^2 + 0.4^2} \cdot \exp\left(j \arctan\left(\frac{0.4}{0.8}\right)\right) = 0.894427 \cdot \exp(j26.565^\circ). \end{aligned}$$

Step 2: Equivalent impedance seen from node 2

Node 2 sees $\underline{Z}_2$ in parallel with the series path $R_3 + \underline{Z}_3$:

$$\underline{Z}_{eq2} = \frac{\underline{Z}_2(R_3 + \underline{Z}_3)}{\underline{Z}_2 + (R_3 + \underline{Z}_3)}.$$

With $\underline{Z}_2 = j30$:

$$\begin{aligned} R_3 + \underline{Z}_3 &= 30 + j60, \\ \underline{Z}_{eq2} &= \frac{j30(30 + j60)}{j30 + (30 + j60)} = \frac{j900 - 1800}{30 + j90} = \frac{27000 + j189000}{9000} = 3 + j21. \end{aligned}$$

Now apply the divider between R_2 and $\underline{Z}_{eq2}$:

$$\begin{aligned} \underline{v}_2 &= \underline{v}_1 \cdot \frac{\underline{Z}_{eq2}}{R_2 + \underline{Z}_{eq2}}. \\ \frac{\underline{v}_2}{\underline{v}_1} &= \frac{3 + j21}{40 + 3 + j21} = \frac{(3 + j21)(43 - j21)}{43^2 + 21^2} = \frac{57 + j84}{229} \\ &= \frac{\sqrt{57^2 + 84^2}}{229} \cdot \exp\left(j \arctan\left(\frac{84}{57}\right)\right) = 0.443291 \cdot \exp(j55.840^\circ). \end{aligned}$$

Step 3: Equivalent impedance seen from node 1

Node 1 sees $\underline{Z}_1$ in parallel with the series path $R_2 + \underline{Z}_{eq2}$:

$$\underline{Z}_{eq1} = \frac{\underline{Z}_1(R_2 + \underline{Z}_{eq2})}{\underline{Z}_1 + (R_2 + \underline{Z}_{eq2})}$$

$$\begin{aligned} R_2 + \underline{Z}_{eq2} &= 40 + 3 + j21 = 43 + j21, \\ \underline{Z}_1(R_2 + \underline{Z}_{eq2}) &= j10(43 + j21) = -210 + j430 \\ \underline{Z}_1 + (R_2 + \underline{Z}_{eq2}) &= j10 + (43 + j21) = 43 + j31 \end{aligned}$$

Thus

$$\begin{aligned} \underline{Z}_{eq1} &= \frac{-210 + j430}{43 + j31} = \frac{(-210 + j430)(43 - j31)}{43^2 + 31^2} = \frac{4300 + j25000}{2810} \\ &= \sqrt{(4300^2 + 25000^2)} \cdot \exp(j \arctan(25000/4300)) \cdot \frac{1}{2810} = 9.027439 \cdot \exp(j80.241^\circ). \end{aligned}$$

Then the divider between R_1 and $\underline{Z}_{eq1}$ gives:

$$\underline{v}_1 = \underline{v}_i \cdot \frac{\underline{Z}_{eq1}}{R_1 + \underline{Z}_{eq1}}.$$

$$\begin{aligned} R_1 + \underline{Z}_{eq1} &= 20 + \frac{4300 + j25000}{2810} = \frac{20 \cdot 2810 + 4300 + j25000}{2810} = \frac{60500 + j25000}{2810} \\ &= \sqrt{(60500^2 + 25000^2)} \cdot \exp(j \arctan(25000/60500)) \cdot \frac{1}{2810} = 23.296022 \cdot \exp(j22.452^\circ) \\ \frac{\underline{v}_1}{\underline{v}_i} &= \frac{9.027439}{23.296022} \cdot \exp(j(80.241^\circ - 22.452^\circ)) = 0.387510 \cdot \exp(j57.789^\circ) \end{aligned}$$

Step 4: Combine the three ratios

Output relative to input:

$$\begin{aligned} \frac{\underline{v}_o}{\underline{v}_i} &= \frac{\underline{v}_3}{\underline{v}_i} = \frac{\underline{v}_3}{\underline{v}_2} \cdot \frac{\underline{v}_2}{\underline{v}_1} \cdot \frac{\underline{v}_1}{\underline{v}_i} \\ &= 0.894427 \cdot \exp(j26.565^\circ) \cdot 0.443291 \cdot \exp(j55.840^\circ) \cdot 0.387510 \cdot \exp(j57.789^\circ) \\ &= 0.153644 \cdot \exp(j140.194^\circ) \\ &\Rightarrow \varphi = 140.2^\circ \end{aligned}$$

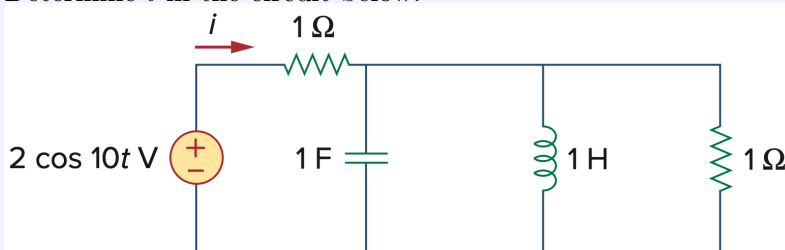
b) Since the phase shift is positive (+140.2°), the output phasor ****leads**** the input phasor.

c) Magnitude of the output when $V_i = 120$ V:

$$V_o = 0.153644 \cdot 120 \text{ V} \approx 18.44 \text{ V}$$

Exercise 5: Nodal Analysis

Determine i in the circuit below.



Solution

$$v_s(t) = 2 \cos(10t) \Rightarrow \omega = 10$$

Capacitor and inductor admittances (1 F and 1 H):

$$\underline{Y}_C = j\omega C = j10, \quad \underline{Y}_L = \frac{1}{j\omega L} = -\frac{j}{10} = -j0.1,$$

and the parallel resistor gives $R = 1$. Thus the total admittance of the parallel block is

$$\underline{Y} = R + \underline{Y}_C + \underline{Y}_L = 1 + j10 - j0.1 = 1 + j9.9.$$

KCL at the node (current through series 1Ω equals current into the parallel block):

$$\frac{\hat{v}_s - \hat{v}}{1} = \hat{v} \underline{Y} \implies \hat{v} (1 + \underline{Y}) = \hat{v}_s \implies \hat{v} = \frac{\hat{v}_s}{1 + \underline{Y}}.$$

The source current $\hat{\underline{i}}$ (which is i in time domain) is the current through the series 1Ω :

$$\hat{\underline{i}} = \frac{\hat{v}_s - \hat{v}}{1} = \hat{v}_s \frac{\underline{Y}}{1 + \underline{Y}}$$

Substitute $\underline{Y} = 1 + j9.9$:

$$\hat{\underline{i}} = 2 \cdot \frac{1 + j9.9}{2 + j9.9} = \frac{200.02 + j19.8}{102.01} \approx (1.9608 + j0.1941) \text{ A}$$

Magnitude, phase and back to the time domain (cosine reference):

$$i(t) = 1.9704 \cos(10t + 5.655^\circ) \text{ A}$$